SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I, K Suthesika (192511070) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K Suthesika

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1)A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

1. Understanding the Given Network – 192.168.20.0/24

Before dividing the network, we first need to understand what **/24** means.

An IPv4 address contains **32 bits**.

The /24 means:

- **24 bits → Network portion**
- **Remaining 8 bits → Host portion**

Therefore: 32 – 24 = 8

Now, the number of possible IP addresses is: $2^8 = 256$

So the original network contains **256 IP addresses**.

The last octet can therefore range from **0 to 255**.

Hence, the complete network range is: **192.168.20.0 – 192.168.20.255**

So, we can think of the original network as one large block containing 256 addresses.

2. Understanding /26, /27 and /28

The subnet masks given for the laboratories are different. The larger the /number, the fewer host addresses are available.

The formula used is: Host bits = 32 – prefix

For Lab A – /26 → 32 – 26 = 6. Lab A has **6 host bits.**

$2^6 = 64$. Lab A has **64 total addresses**.

For Lab B – /27 →32 – 27 = 5. Lab B has **5 host bits.**

$2^5 = 32$. Lab B has **32 total addresses.**

For Lab C – /28 → 32 – 28 = 4**.** Lab C has **4 host bits.**

$2^4 = 16$. Lab C has **16 total addresses.**

3. Finding Usable Host Addresses

In every normal subnet, two addresses are reserved:

- The **first address** is the Network Address.
- The **last address** is the Broadcast Address.

Therefore: Usable hosts = Total addresses – 2

Lab A → 64 – 2 = 62 usable hosts

Lab B → 32 – 2 =30 usable hosts

Lab C → 16 – 2 = 14 usable hosts

4. Finding the Subnet for Lab A – /26

Lab A is assigned: **192.168.20.0/26**

We already know that /26 provides **64 addresses**.

Lab A starts at: **192.168.20.0**

To find where this subnet ends, we add 64: 0+64=64

The next subnet would begin at .64.

Therefore, the current subnet ends one address before .64: 64-1=63

So Lab A occupies: **192.168.20.0 – 192.168.20.63**

Now identify the three important parts:

Network Address

The **first address** is always the network address: **192.168.20.0**

Broadcast Address

The **last address** is the broadcast address: **192.168.20.63**

Valid Host Range

The addresses between the network and broadcast addresses can be assigned to devices:

192.168.20.1 – 192.168.20.62

Therefore:

Lab A	Address
Network Address	192.168.20.0
Valid Host Range	192.168.20.1 – 192.168.20.62
Broadcast Address	192.168.20.63
Usable Hosts	62

5. Finding the Subnet for Lab B – /27

Lab A has already used: **192.168.20.0 – 192.168.20.63**

Therefore, the **next available address** is: **192.168.20.64**

Lab B uses /27.

We calculated that /27 provides **32 total addresses**.

So we start at .64 and add 32: 64+32=96

The next subnet would start at .96.

Therefore, Lab B ends at: 96-1=95

So Lab B occupies: **192.168.20.64 – 192.168.20.95**

Now identify the addresses:

Network Address → First address: **192.168.20.64**

Broadcast Address → Last address: **192.168.20.95**

Valid Host Range → Addresses between them: **192.168.20.65 – 192.168.20.94**

Therefore:

Lab B	Address
Network Address	192.168.20.64
Valid Host Range	192.168.20.65 – 192.168.20.94
Broadcast Address	192.168.20.95
Usable Hosts	30

6. Finding the Subnet for Lab C – /28

Lab B has used: **192.168.20.64 – 192.168.20.95**

Therefore, the next available address is: **192.168.20.96**

Lab C uses /28.

For /28: 32-28=4

So there are **4 host bits**.

Therefore: $2^4 = 16$

Lab C has **16 total addresses**.

Starting from .96: 96+16=112

The next subnet would begin at .112.

Therefore, Lab C ends at: 112-1=111

So Lab C occupies: **192.168.20.96 – 192.168.20.111**

Now identify the addresses:

Network Address : 192.168.20.96

Broadcast Address: 192.168.20.111

Valid Host Range: 192.168.20.97 – 192.168.20.110

Therefore:

Lab C	Address
Network Address	192.168.20.96
Valid Host Range	192.168.20.97 – 192.168.20.110
Broadcast Address	192.168.20.111
Usable Hosts	14

Final Subnetting Table

Lab	CIDR	Total Addresses	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
A	/26	64	192.168.20.0	192.168.20.1 – 192.168.20.62	192.168.20.63	62
B	/27	32	192.168.20.64	192.168.20.65 – 192.168.20.94	192.168.20.95	30
C	/28	16	192.168.20.96	192.168.20.97 – 192.168.20.110	192.168.20.111	14

Total Usable Host Addresses : 62 + 30 + 14 = 106

2)A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

1. Understanding the Given Subnet Mask

Both systems have the same subnet mask: **255.255.255.192**

First, we convert this subnet mask into CIDR notation.

The last octet is: **192**

In binary: 192=11000000

Therefore, the complete subnet mask is:

```
255       . 255       . 255       . 192
11111111  . 11111111  . 11111111  . 11000000
```

There are: 8+8+8+2=26 network bits.

Therefore: 255.255.255.192=/26

So, both systems are using a **/26 subnet**.

2. Finding the Number of Addresses

For a /26 subnet: Host bits = 32 – 26 = 6

Therefore, the total number of addresses in one subnet is: $2^4 = 16$

So, each /26 subnet contains: 64 total IP addresses

Out of these 64 addresses:

- First address → Network address
- Last address → Broadcast address

Therefore, usable host addresses are: 64 – 2 = 62

Hence: 62 usable host addresses.

3. Finding the Subnet Ranges

To find the subnet ranges, we calculate the **block size**.

The last octet of the subnet mask is **192**.

Therefore: Block size = 256 – 192 = 64

So, the subnet ranges increase by **64**.

Starting from 0:

0

64

128

192

Therefore, the possible /26 subnet ranges are:

Subnet 1: 192.168.1.0 – 192.168.1.63

Subnet 2: 192.168.1.64 – 192.168.1.127

Subnet 3: 192.168.1.128 – 192.168.1.191

Subnet 4: 192.168.1.192 – 192.168.1.255

Now we can check where each system's IP address belongs.

4. Finding the Subnet for System 1

System 1 has the IP address: **192.168.1.130**

From the subnet ranges above:

192.168.1.0 – 192.168.1.63

192.168.1.64 – 192.168.1.127

192.168.1.128 – 192.168.1.191

192.168.1.192 – 192.168.1.255

The value **130** falls between **128 and 191**.

Therefore, System 1 belongs to: 192.168.1.128/26

Network Address of System 1

The first address of the subnet is the network address → 192.168.1.128

Broadcast Address of System 1

The last address of the subnet is the broadcast address → 192.168.1.191

Valid Host Range of System 1

The addresses between the network and broadcast addresses can be assigned to hosts.
192.168.1.129-192.168.1.190

Checking System 1 IP

Given IP: **192.168.1.130**

Valid host range: **192.168.1.129 – 192.168.1.190**

Since: It falls between: **192.168.1.129 – 192.168.1.190**, the given IP belongs to the valid host range.

Therefore: System 1 IP is a valid host address

System 1 Result

Item	Result
IP Address	192.168.1.130
Subnet Mask	255.255.255.192 (/26)
Subnet Range	192.168.1.128 – 192.168.1.191
Network Address	192.168.1.128
Valid Host Range	192.168.1.129 – 192.168.1.190
Broadcast Address	192.168.1.191
Usable Hosts	62
IP Validity	Valid

5. Finding the Subnet for System 2

System 2 has the IP address: **192.168.1.190**

Again, compare it with the subnet ranges:

192.168.1.0 – 192.168.1.63

192.168.1.64 – 192.168.1.127

192.168.1.128 – 192.168.1.191

192.168.1.192 – 192.168.1.255

The value **190** also falls between **128 and 191**.

Therefore, System 2 belongs to: 192.168.1.128/26

Network Address of System 2

The first address of this subnet is: 192.168.1.128

Broadcast Address of System 2

The last address of this subnet is: 192.168.1.191

Valid Host Range of System 2

The usable addresses are: 192.168.1.129 – 192.168.1.190

Checking System 2 IP

Given IP: **192.168.1.190**

Valid host range: **192.168.1.129 – 192.168.1.190**

Since: It falls between: **192.168.1.129 – 192.168.1.190**, the given IP belongs to the valid host range.

Therefore: System 2 IP is a valid host address

System 2 Result

Item	Result
IP Address	192.168.1.190
Subnet Mask	255.255.255.192 (/26)
Subnet Range	192.168.1.128 – 192.168.1.191
Network Address	192.168.1.128
Valid Host Range	192.168.1.129 – 192.168.1.190
Broadcast Address	192.168.1.191

Usable Hosts	62
IP Validity	Valid

Final Subnetting Table

System	IP Address	Subnet Range	Network Address	Valid Host Range	Broadcast Address	Usable Hosts	Valid IP?
System 1	192.168.1.130	192.168.1.128 – 192.168.1.191	192.168.1.128	192.168.1.129 – 192.168.1.190	192.168.1.191	62	Yes
System 2	192.168.1.190	192.168.1.128 – 192.168.1.191	192.168.1.128	192.168.1.129 – 192.168.1.190	192.168.1.191	62	Yes

3)An organization has been allocated the IP block 172.16.0.0/16 for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

1. Understanding the Given Network – 172.16.0.0/16

The organization has: **172.16.0.0/16**

An IPv4 address contains **32 bits**.

The /16 means:

- First **16 bits → Network portion**
- Remaining **16 bits → Host portion**

Therefore : Host bits = 32 -16 = 16

The total number of addresses in the original network is: $2^{16} = 65{,}536$

Therefore, the original network contains addresses from: 172.16.0.0 to 172.16.255.255
So, this is one large network block containing **65,536 IP addresses**.

2. Understanding the Department Subnets

Each department has been given a different subnet size.

The important formula is: Host bits = 32 - prefix

Then: Total addresses = $2^{\text{Host bits}}$

And: Usable hosts = Total addresses -2

Let's calculate the size required by each department.

HR → /25

Host bits: 32-25=7

Total addresses: $2^7 = 128$

Usable hosts: 128 – 2 = 126

Therefore: **HR needs 128 total addresses and has 126 usable host addresses.**

Sales → /26

Host bits: 32 – 26 = 6

Total addresses: $2^6 = 64$

Usable hosts: 64 – 2 = 60

Therefore: **Sales needs 64 total addresses and has 62 usable host addresses.**

IT → /27

Host bits: 32 – 27 = 5

Total addresses: $2^5 = 32$

Usable hosts: 32 – 2 = 30

Therefore: **IT needs 32 total addresses and has 30 usable host addresses.**

Admin → /28

Host bits: 32 – 28 = 4

Total addresses: $2^4 = 16$

Usable hosts: 16 – 2 = 14

Therefore: **Admin needs 16 total addresses and has 14 usable host addresses.**

3. Assigning the Subnet to HR – /25

We start from the beginning of the organization's network: **172.16.0.0**

HR requires /25.

We already calculated that /25 contains: 128 addresses

So HR starts at: **172.16.0.0**

To find where it ends: 0 + 128 = 128

The next subnet would start at .128.

Therefore, HR ends one address before .128: 128 – 1 = 127

So HR occupies: **172.16.0.0 – 172.16.0.127**

Network Address

The first address: 172.16.0.0

Broadcast Address

The last address: 172.16.0.127

Valid Host Range

The addresses between them are: 172.16.0.1-172.16.0.126

HR Result

Item	Address
Subnet	/25
Network Address	172.16.0.0
Valid Host Range	172.16.0.1 – 172.16.0.126
Broadcast Address	172.16.0.127
Usable Hosts	126

4. Assigning the Subnet to Sales – /26

HR has already used: **172.16.0.0 – 172.16.0.127**

Therefore, the **next available address** is: **172.16.0.128**

Sales is assigned /26.

A /26 subnet contains: $2^6 = 64$

So Sales needs **64 addresses**.

Starting from .128: 128 + 64 = 192

Therefore, the next subnet would start at .192.

So Sales ends at: 192 – 1 =190

Hence, Sales occupies: **172.16.0.128 – 172.16.0.191**

Network Address: 172.16.0.128

Broadcast Address: 172.16.0.191

Valid Host Range: 172.16.0.129-172.16.0.190

Sales Result

Item	Address
Subnet	/26
Network Address	172.16.0.128
Valid Host Range	172.16.0.129 – 172.16.0.190
Broadcast Address	172.16.0.191
Usable Hosts	62

5. Assigning the Subnet to IT – /27

Sales has used: **172.16.0.128 – 172.16.0.191**

Therefore, the next available address is: **172.16.0.192**

IT is assigned /27.

A /27 subnet contains: $2^5 = 32$

So IT needs **32 addresses**.

Starting from .192: 192 + 32 = 224

Therefore, the next subnet would start at .224.

So IT ends at: 224-1=223

Hence, IT occupies: **172.16.0.192 – 172.16.0.223**

Network Address: 172.16.0.192

Broadcast Address: 172.16.0.223

Valid Host Range: 172.16.0.193-172.16.0.222

IT Result

Item	Address
Subnet	/27
Network Address	172.16.0.192

Valid Host Range	172.16.0.193 – 172.16.0.222
Broadcast Address	172.16.0.223
Usable Hosts	30

6. Assigning the Subnet to Admin – /28

IT has used: **172.16.0.192 – 172.16.0.223**

Therefore, the next available address is: **172.16.0.224**

Admin is assigned /28.

A /28 subnet contains: $2^4 = 16$

So Admin needs **16 addresses**.

Starting from .224: 224+16=240

Therefore, the next subnet would start at .240.

So Admin ends at: 240-1=239

Hence, Admin occupies: **172.16.0.224 – 172.16.0.239**

Network Address: 172.16.0.224

Broadcast Address: 172.16.0.239

Valid Host Range: 172.16.0.225-172.16.0.238

Admin Result

Item	Address
Subnet	/28
Network Address	172.16.0.224
Valid Host Range	172.16.0.225 – 172.16.0.238
Broadcast Address	172.16.0.239
Usable Hosts	14

7. Finding the Remaining Unused IP Range

This is the final part of the question.

The Admin subnet ends at: **172.16.0.239**

The original network continues all the way to: **172.16.255.255**

Therefore, the remaining unused addresses start from: 172.16.0.240 and continue up to: 172.16.255.255

So the remaining unused IP range is: 172.16.0.240 – 172.16.255.255

Important Note

This range is the **remaining portion of the original /16 block** after the four requested subnets have been allocated. It is not one single /26, /27, or /28 subnet; it is simply the unused portion of the organization's original address space.

Final Subnetting Table

Department	CIDR	Total Addresses	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
HR	/25	128	172.16.0.0	172.16.0.1 – 172.16.0.126	172.16.0.127	**126**
Sales	/26	64	172.16.0.128	172.16.0.129 – 172.16.0.190	172.16.0.191	**62**
IT	/27	32	172.16.0.192	172.16.0.193 – 172.16.0.222	172.16.0.223	**30**
Admin	/28	16	172.16.0.224	172.16.0.225 – 172.16.0.238	172.16.0.239	**14**

4)A company uses the IP address 172.16.10.0/24 and has implemented subnetting using a /27 subnet mask for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address 172.16.10.78 belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address 172.16.10.95 falls within the same subnet range based on the given configuration.

Subnetting of 172.16.10.0/24 using /27

Given:

- Network: **172.16.10.0/24**
- Subnet mask: **/27 = 255.255.255.224**

1. Number of Subnets

Bits borrowed: 27 – 24 = 3 → $2^3 = 8$ subnets

2. Usable Hosts per Subnet

Host bits: 32 – 27 = 5

Total addresses: $2^5 = 32$

Usable hosts: 32 – 2 = 30

3. Subnet Containing 172.16.10.78

Block size: 256 – 224 = 32

Subnets are:

- 0–31
- 32–63
- **64–95**
- 96–127
- ...

Since **78** lies between **64 and 95**: 172.16.10.64/27

- **Network Address:** 172.16.10.64
- **Valid Host Range:** 172.16.10.65 – 172.16.10.94
- **Broadcast Address:** 172.16.10.95

4. Check 172.16.10.95

172.16.10.95 is within the subnet range **172.16.10.64–95**, but it is the **broadcast address**.

Therefore:

- **172.16.10.78 → Valid host IP ✅**
- **172.16.10.95 → Same subnet, but broadcast address ❌ for host assignment**

Final Answer

Requirement	Answer
Number of subnets	**8**
Usable hosts/subnet	**30**
Subnet of .78	**172.16.10.64/27**
Network	**172.16.10.64**
Host range	**172.16.10.65 – 172.16.10.94**
Broadcast	**172.16.10.95**
.78 valid host?	**Yes**
.95 same subnet?	**Yes, but it is broadcast**

5)A network uses the Internet Protocol version 6 address 2001:0db8:0000:0000:0000:ff00:0042:8329. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length. si jA network uses the Internet Protocol version 6 address 2001:0db8:0000:0000:0000:ff00:0042:8329. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

IPv6 Addressing

Given:

IPv6 Address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Prefix length: **/64**

1. Compress the IPv6 Address

IPv6 rules:

- Remove leading zeros from each block.
- Replace the longest consecutive sequence of 0000 blocks with :: .

Given:

2001:0db8:0000:0000:0000:ff00:0042:8329

Compressed form: 2001:db8::ff00:42:8329

2. Expand the Compressed Address

Compressed: **2001:db8::ff00:42:8329**

The :: represents three 0000 blocks.

Therefore: 2001:0db8:0000:0000:0000:ff00:0042:8329

3. Find the Network Prefix

Given prefix: **/64**

The first **64 bits** (first 4 groups) represent the network.

Therefore: Network Prefix = 2001:0db8:0000:0000::/64 or simply: 2001:db8:0:0::/64

4. Number of Possible Host Addresses

IPv6 has **128 bits**.

With /64: 128 – 64 = 64 host bits remain.

Therefore: 2^{64}

possible host addresses.

$$2^{64} = 18,446,774,073,709,551,616$$

Final Answer

Requirement	**Answer**
Full IPv6 Address	**2001:0db8:0000:0000:0000:ff00:0042:8329**
Compressed Address	**2001:db8::ff00:42:8329**
Expanded Address	**2001:0db8:0000:0000:0000:ff00:0042:8329**
Network Prefix	**2001:db8:0:0::/64**
Host Bits	**64 bits**
Possible Host Addresses	**2^{64} = 18,446,744,073,709,551,616**